



Chemistry 1 Honors

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 9–12 · FULL YEAR · HONORS

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. More investigations and examples are approved for this course than appear here, so scholars may see some variation in labs, activities, and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Chemistry is the study of matter and how it changes, and Chemistry 1 Honors explores it with the depth and pace an honors course demands. Scholars work through structured modules built on interactive videos, simulations, and immersive lab experiences on Optima's VR campus, keeping a digital science notebook as they go. Laboratory investigation is integral, not decorative: scholars conduct digital and VR experiments, collect and interpret data, and write lab reports throughout the year. The journey runs from atoms and the periodic table through bonding, reactions, gases, equilibrium, and acids and bases to electrochemistry and nuclear chemistry.

THE YEAR AT A GLANCE

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| 0 Lab Safety and Course Launch (start of year)
Lab safety contract, course routines, and setting up the digital science notebook. | 1 Intro to Chemistry and Measurement (Semester 1)
The science of matter, states of matter, and the measurement and unit skills every later module leans on. |
| 2 Atomic Structure and the Periodic Table (Semester 1)
How the atomic model evolved, the structure of the atom, and what the periodic table predicts. | 3 Chemical Bonds and Molecular Geometry (Semester 1)
Ionic, covalent, and metallic bonding and the shapes molecules take. |
| 4 Chemical Reactions and Stoichiometry (Semester 1)
Balancing equations, reaction types, and the mole concept; the semester closes with a midterm exam. | 5 Gas Laws and Thermochemistry (Semester 2)
Kinetic molecular theory, gas behavior, and energy in chemical systems. |
| 6 Liquids, Solids, Solutions, and Colloids (Semester 2)
Phase changes, solution chemistry, and the special properties of water. | 7 Thermodynamics, Equilibrium, and Acid-Base Chemistry (Semester 2)
Reaction rates, dynamic equilibrium, pH, and acid-base behavior. |
| 8 Electrochemistry and Redox Reactions (Semester 2)
Oxidation and reduction in living and non-living systems. | 9 Nuclear Chemistry and Its Applications (Semester 2)
Radioactive decay, fission and fusion, and real-world applications; the year closes with a final exam. |

A TYPICAL WEEK

- Weekly Canvas lessons with checkpoints and quizzes; a recommended pace of two to three lessons per week.
- A digital science notebook tracks learning all year, with vocabulary builders and Apply Your Knowledge tasks.
- Labs on Optima's VR campus, with video walk-through alternatives so scholars without a headset complete every lab.
- Optional virtual office hours for help and lab walk-throughs.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Lessons, notebook work, and labs fill most of the scheduled time.
- Each semester grade weighs the two quarters at 40% each plus a midterm or final exam at 20%.

WHAT SCHOLARS NEED

- A working computer with reliable internet and access to Canvas.
- A digital science notebook (OneNote or similar).
- A VR headset for immersive labs — optional, with video walk-through alternatives provided.
- Microsoft 365 and OneDrive (provided to every scholar) for lab reports and presentations.

Grading at a glance: Assessments (quizzes and exams) 36% · Assignments 34% · Labs and VR 30%

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.